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Housing with a housing part comprising an integrated protective section

Abstract

In order to improve a housing, in particular for an electrotechnical apparatus, comprising at least two housing parts which each have a connecting face, said connecting faces, in a closed state of the housing, facing one another along a connecting section, it is proposed that at least one of the at least two housing parts, along the connecting section, has at least one integrated protective section, in particular for EMC shielding and/or for mechanical protection.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION (1) This application is a continuation of international application number PCT/EP2021/057831 filed on Mar. 25, 2021, and claims the benefit of German application No. 10 2020 109 898.7 filed on Apr. 8, 2020. (2) This patent application relates to the subject matter disclosed in and claims the benefit of international application PCT/EP2021/057831, filed Mar. 25, 2021, and German application No. 10 2020 109 898.7, filed Apr. 8, 2020, the teachings and disclosure of which are hereby incorporated in their entirety by reference thereto for all purposes.

BACKGROUND OF THE INVENTION

(1) The invention relates to a housing, in particular for an electrotechnical apparatus, the housing comprising at least two housing parts and the at least two housing parts each having a connecting face, said connecting faces, in a closed state of the housing, facing one another along a connecting section.

(2) The invention additionally relates to an electrotechnical apparatus and to a motor vehicle component and to a motor vehicle, each in particular comprising a housing of the aforementioned kind.

(3) In particular, an interior of the housing, which is surrounded at least in part by the at least two housing parts, is configured as a receptacle for electrical components of the electrotechnical apparatus. In particular, electrical components of the electrotechnical apparatus are arranged in the interior when in an assembled state. For example, at least one electrical component is a car battery and/or an electrical control device.

(4) One object forming the basis of the invention is that of improving a housing and/or an electrotechnical apparatus and/or a motor vehicle component and/or a motor vehicle, in particular to improve the functionality thereof and/or to increase the operational safety thereof.

SUMMARY OF THE INVENTION

(5) In some embodiments of the invention, the above-mentioned object is achieved by a housing which comprises at least two housing parts and the at least two housing parts each having a connecting face, said connecting faces, in a closed state of the housing, facing one another along a connecting section and at least one of the at least two housing parts has at least one integrated protective section.

(6) For example, at least one integrated protective section serves for mechanical protection, in particular against fluids, preferably against water and/or cleaning liquids, in particular from a high-pressure cleaner, and/or against small solid particles, in particular dust particles, abrasive particles and/or dirt particles.

(7) In particular, at least one integrated protective section is configured for electromagnetic compatibility (EMC) shielding. Preferably, this integrated protective section reduces the propagation of electromagnetic waves from the interior of the housing to the outside and/or from the outside into the interior, and, in particular, such propagation is at least approximately prevented by the protective section. In particular, in at least one protective section, the configuration for EMC protection is possible alternatively or additionally to the configuration as mechanical protection.

(8) In particular, one of the advantages of embodiments of the invention is thus considered to lie in the fact that the interior of the housing is better protected by the integrated protective section, in particular from external foreign influences, for example from external material solid and/or liquid contaminants and/or electromagnetic influences, and in particular electrical components arranged in the housing interior are also protected.

(9) Preferably, interference originating from electrical components arranged in the interior of the housing and affecting the area surrounding the housing, for example further parts of the electronic

apparatus or of the motor vehicle, is also at least reduced or even at least approximately prevented by at least one protective section configured in particular as an EMC shielding section.
(10) In particular, the protective section configured for EMC shielding is configured to hinder the propagation of electromagnetic waves and thus increases the electromagnetic compatibility (EMC) of the housing with, in particular, electrical components arranged therein. Preferably, the protective section is configured here in such a way that it fulfils at least one, in particular several, for example all requirements of corresponding standards and/or legal regulations in respect of electromagnetic compatibility.
(11) For example, one advantage of embodiments of the invention is considered to lie in the fact that at least one of the housing parts has at least one integrated protective section, and in particular the integration of the protective section on the housing part simplifies an assembly of the latter, and therefore, for example, the housing can be produced more cost-effectively.
(12) In some advantageous embodiments, it is provided that exactly one protective section is provided along the connecting section. For example, this allows the corresponding housing part to be produced simply and thus cost-effectively.
(13) In other particularly favourable embodiments, it is provided that exactly two protective sections are provided along the connecting section, whereby, for example, these protective sections can be configured in a correspondingly specialised manner for a particular requirement each, in particular on the one hand for mechanical protection and on the other hand for EMC shielding, and thus preferably a protective effect can be increased.
(14) In some advantageous embodiments with a plurality of protective sections along the connecting section, both of the at least two housing parts each have at least one integrated protective section.
(15) In other particularly advantageous embodiments, it is provided that in the case of a plurality of protective sections provided, only one of the at least two housing parts has the plurality of protective sections, whereby in particular the production of the other of the two housing parts is simpler and thus more cost-effective.
(16) With regard to the configuration of the at least one protective section and/or the plurality of protective sections, no further details have yet been provided.
(17) In particular, one or the exactly one protective section and/or a plurality, for example all, of the plurality of, in particular the exactly two, protective sections have one or more of the features explained below. Thus, when a feature is explained below in conjunction with at least one protective section, this formulation is to be understood in such a way that the exactly one protective section and/or one, preferably a plurality, for example all, of the plurality of, in particular the exactly two, protective sections have this explained feature.
(18) It is particularly favourable if at least one protective section is arranged captively on one of the housing parts. In particular, this means that the at least one protective section is fixedly connected to one of the housing parts and is integrated into same, and preferably that assembly is thus simplified and less prone to errors, since fewer individual components requiring assembly are provided.
(19) In particularly advantageous embodiments, it is provided that a section of the one housing part forms at least one protective section. In particular, no separate component is thus required for the at least one protective section, so that in particular production costs can be reduced and preferably the assembly of the housing can be simplified.
(20) It is particularly favourable if at least one protective section is a transformed section of the one housing part, thereby allowing simple production and formation thereof.
(21) For example, at least one protective section, in particular the at least one transformed protective section, is pressed or stamped into the one housing part.
(22) In some preferred embodiments, it is provided that at least one protective section is a section integrally formed on the one housing part.

(23) For example, the integrally formed section is formed from a different material than the housing part. In other advantageous embodiments, the integrally formed section is made of the same material as the one housing part.
(24) In particular, it is possible in the case of the integrally formed section to form it in accordance with special requirements and preferably to connect it fixedly to the one housing part, so that in particular by means of the integrally formed section an integrated protective section which can be formed flexibly and can be connected well to the housing part can be produced in a favourable manner.
(25) In particularly advantageous embodiments, it is provided that a protective element associated with a housing part forms at least one protective section. In particular, if the protective element is a prefabricated element different from the housing part, these two can be conceived and manufactured in accordance with their requirements and, in particular, due to their integration, for example as explained below, the one housing part can be provided with an integrated protective section prior to the assembly of the housing, and said assembly can be performed simply and in a manner less prone to errors.
(26) In particular, the protective element is fixed to the associated housing part, in particular already prior to assembly.
(27) It is particularly advantageous if the protective element is connected to the associated housing part with a substance-to-substance bond, in a positively-locking manner and/or in a force-locking manner.
(28) In particular, the protective element is arranged with a retaining section, in particular along the connecting section at least partially, preferably continuously, on the associated housing part, in particular captively.
(29) In particular, the protective element, in particular a retaining section thereof, is connected to the one housing part by means of clinching.
(30) In particularly advantageous embodiments, it is provided that the protective element is connected to the one housing part by a substance-to-substance bond, for example by welding.
(31) In different embodiments, the protective element is produced from different materials.
(32) In particularly favourable embodiments, it is provided that the protective element is formed from a metal material. For example, this allows it to be produced favourably and stably. The conductivity of the metal material preferably increases the EMC shielding.
(33) For example, the metal material comprises steel, in particular carbon steel or spring steel.
(34) In some favourable embodiments, the metal material comprises aluminium and/or copper and/or beryllium.
(35) For example, the metal material is an alloy, in particular comprising copper and/or beryllium and/or aluminium.
(36) Preferably, the protective element is coated, for example with a metal coating, in particular one comprising nickel and/or copper.
(37) In some advantageous embodiments, the protective element is made of, in particular, nickel- or copper-plated steel, for example spring steel or cold-rolled strip.
(38) Preferably, the protective element is produced from a flat material. It is particularly favourable if the protective element is a metal layer, for example made of sheet metal. Preferably, the protective element is formed as a ring which frames a housing part opening of the oppositely arranged housing part in a closed manner circumferentially.
(39) For example, such an embodiment can be produced favourably and can be integrated in a space-saving manner, so that a protective section with a good protective function can thus be realised cost-effectively and also with, at most, a small free installation space.
(40) It is particularly advantageous if at least one protective section is a transformed section of the protective element, in particular of the, for example, metal layer. In particular, this makes it possible to realise a structurally simpler protective section that can be produced cost-effectively.

(41) It is particularly favourable if at least one protective section is flat.
(42) In particular, at least one flat protective section has a contacting face with which the oppositely arranged housing part is contacted at least in the closed state. In particular, the contacting face allows good contact, preferably over the entire surface, with the opposite housing part, which in particular improves EMC shielding.
(43) Alternatively or additionally, it is favourable if at least one protective section is formed flat to cover a gap between the connecting faces in the closed state of the housing, whereby the gap is covered along the protective section and mechanical protection against external influences is improved.
(44) In some favourable embodiments, it is provided that at least one protective section has an edge. In particular, the edge has a spike-like cross-section transverse to a longitudinal extent of the protective section, the longitudinal extent being directed in particular along the connection contour. The edge favourably protrudes beyond a connecting side of the housing part that has this protective section. It is particularly advantageous if the edge is configured to extend toward the opposite housing part, in particular up to an edge tip, in particular starting from the housing part or starting from the protective element. Preferably, the edge contacts the opposite housing part, for example with its edge tip, and is in particular pressed onto the opposite housing part. This allows, for example, good contact between the housing parts or the opposite housing part and the protective element. In some embodiments, a fixed connection between the housing parts, for example, is also realised by means of the edge.
(45) In some advantageous embodiments, at least one protective section is formed by a bead. For example, the bead is a half bead or a full bead. The configuration of the beads allows for a wide range of possibilities, whereby the protective section can be configured in a structurally simple manner in accordance with specific requirements.
(46) In some favourable embodiments, the bead along the connecting section has substantially the same shape and/or dimensions.
(47) In some particularly advantageous embodiments, the shape and/or dimensions of the bead vary along the connecting section. For example, the height of the bead varies along the connecting section, preferably being shorter at fastening points and/or points where loading occurs than between these points, which in particular makes contact with the opposite housing part more uniform and improved. In particular, the bead height is measured as the spacing between mutually averted sides of a bead top and a bead foot in the case of a full bead and between mutually averted sides of adjacent bead feet in the case of a half bead. In particular, the bead is arranged with two bead feet on spaced-apart sections, and in the case of a full bead the projecting bead top is arranged between the two bead feet and in the case of a half bead the two bead feet are arranged offset transversely to the bead course and projecting away from each other.
(48) For example, the bead forming the at least one protective section is formed in one of the housing parts. In advantageous embodiments, the bead forming the at least one protective section is provided in the protective element, which is in particular formed as a metal layer.
(49) In particularly favourable embodiments, at least one protective section is spring-mounted. In particular, this improves contacting of the oppositely arranged housing part, since tolerances of the other housing part can be compensated for by the spring-mounted configuration.
(50) For example, at least one protective section is resiliently connected to the rest of the housing part or protective element forming this protective section.
(51) Preferably, the at least one resiliently connected protective section is resiliently connected to a section of the housing part forming this protective section, said section of the housing part forming the connecting face. Alternatively or additionally, it is provided, for example, that this protective section is resiliently connected to a cover section which covers a housing part opening of the oppositely arranged housing part in the closed state.
(52) When the resiliently connected protective section is formed by the protective element, it is

particularly favourable if this protective section is resiliently connected to the retaining section of the protective element.
(53) With regard to the further arrangement of the at least one protective section, in particular with regard to the arrangement with respect to the opposite housing part, no further details have yet been provided.
(54) It is particularly favourable if at least one protective section in the closed state of the housing contacts the oppositely arranged of the at least two housing parts along the connecting section at least partially, preferably at least approximately continuously. This is particularly favourable for EMC shielding, since an at least partially continuous connection between the housing parts is hereby established and the propagation of the electromagnetic waves is thereby preferably at least reduced. The at least partial contact can also be favourable for mechanical protection, since the gap is at least substantially closed at these points, for example, and thus contaminants in particular cannot enter the gap.
(55) In particular, at least one protective section establishes a metallic and/or galvanic contact between the at least two housing parts at least in sections, for example continuously, along the connecting section.
(56) It is favourable if at least one protective section electrically conductively connects the at least two housing parts along the connecting section at least in sections, for example continuously.
(57) In particularly advantageous embodiments, it is provided that at least one protective section, in the closed state of the housing, at least partially abuts the connecting face of the oppositely arranged housing part. In particular, this offers a structurally simple solution and assembly is simplified because the connecting face already faces the housing part with the protective section. In particular, the connecting face is a reworked face and thus has low tolerances, so that contacting is improved.
(58) In some advantageous embodiments, it is provided that at least one protective section in the closed state of the housing at least partially abuts a wall section face of the oppositely arranged housing part, said wall section face extending away from the connecting face. For example, this is favourable in the case of housing parts with thin outer walls, in which little installation space is available at the connecting face, so that this little installation space can be used in particular for a sealing element and/or a further protective section. Abutment against the wall section face is also favourable in particular, so that the gap is continuously covered and thus particularly well protected from external influences.
(59) In some advantageous embodiments, it is provided that at least one protective section along the connecting section is at least partially spaced from the oppositely arranged housing part. This is particularly favourable for an outside protective section, in which case there is still a free space between this protective section and a sealing element arranged in the gap between the connecting faces, it being possible for dirt and/or moisture to collect in this free space and to escape again from the gap due to the clearance provided by the spacing.
(60) In particular, it is favourable if the protective section spaced-apart from the oppositely arranged housing part is spaced from an outer wall section face thereof and, in particular, has an end region which, viewed from the outside, is arranged at least partially in front of the wall section face but spaced therefrom. For example, the protective section is formed here as a collar covering the gap. In particular, this protective section extends from this wall region in the direction of the other housing part on which it is integrated, in particular as explained above, so that this protective section preferably covers the gap between the connecting faces on the outside, in particular for mechanical protection, and is arranged partially overlapping but in particular spaced from the wall section face.
(61) In some advantageous embodiments, the configuration, in particular a shape and/or dimensions, of at least one protective section along the connecting section is substantially the same. In particular, this offers a structurally simple and cost-effective solution.

(62) In some preferred embodiments, the configuration, for example a shape and/or dimensions, of at least one protective section varies along the connecting section, so that preferably the protective effect thereof can be optimised. For example, a height of the protective section measured transversely to the connecting faces varies. In particular, it is greater between points where loading occurs than at these loading points. Preferably, this improves continuous contacting.
(63) In preferred embodiments, at least one protective section is configured to extend continuously along the connecting section. For example, a good and continuous protective effect can be achieved in this way.
(64) In some particularly advantageous embodiments, it is provided that at least one protective section has one or more recesses along the connecting section. This is particularly favourable in order to allow contaminants and/or moisture to pass out of the recess behind the protective section, for example between the latter and a sealing element. Therefore, it is particularly favourable to make the recesses small in order to keep interruption of the protection small, but large enough to allow moisture and/or liquid to escape, although in particular a capillary action at the recess and/or along the protective section should be avoided.
(65) In particular, it is favourable to provide the recesses in the case of a protective section arranged on the outside.
(66) Preferably, one or more recesses are formed in an angled manner, so that in particular this recess does not run directly transversely to the longitudinal extent of the protective section and thus also in the region of the recess there is substantially no interruption of the protection by the protective section.
(67) It is particularly advantageous if the protective section has two partial sections in the region of a recess, which run offset to each other transversely to the connection contour and in particular each run at least approximately parallel to the connection contour. Preferably, the recess runs in the region of the offset course between the two partial sections, so that in particular in this region the recess is delimited by the two partial sections in the direction transverse to the connection contour. In particular, the two partial sections each have, in the region of the recess, an end at which, in particular, the offset course of the two partial sections also ends. It is particularly advantageous here that the recess runs at an angle from the end of one of the partial sections on a transverse side of the protective section relative to the connection contour, at least approximately in the direction of the connection contour between the two partial sections, ultimately to the end of the other partial section on the opposite transverse side. Thus, in the case of such a particularly angled recess, an escape of liquid and/or moisture is favourably enabled, with continuous protection still being provided by the overlapping course of the partial sections, in particular protection for the inwardly arranged sealing element against damaging influences directly acting from the outside, such as impinging cleaning liquid from a high-pressure cleaner.
(68) Preferably, a longitudinal extent of the one or more recesses, in particular along the connection contour, is at most 10 mm, in particular at most 5 mm, for example at most 2 mm long and/or at least 0.1 mm, for example at least 0.5 mm, in particular at least 1 mm long. For example, the longitudinal extent of the recesses is substantially smaller, for example at least five times, in particular at least ten times, smaller than a longitudinal extent of a section of the protective section adjoining the recess. For example, the recesses can be arranged at irregular spacings from one another, and in other advantageous embodiments the recesses are arranged at regular spacings from one another.
(69) In particular, the recesses are suitably configured for EMC shielding, for example their spacing is equal to or an integer multiple of a wavelength of an electromagnetic wave to be shielded.
(70) It is particularly advantageous if at least one protective section is arranged on the inside along the connecting section. In particular, the protective section arranged on the inside is arranged between a sealing element, arranged in the gap between the connecting faces, and the interior. In

particular, an installation space present on the inside is also favourably utilised by the protective section, wherein, for example, a region of the in particular lid-like housing part adjacent to the connecting face can also be used for the formation and/or arrangement of the protective section on the inside, in particular for the forming, for example the stamping, of the protective section or for the connection thereof to the housing part by means of the retaining section.
(71) In particular, the protective section provided on the inside at least partially contacts the housing part arranged opposite.
(72) Preferably, the connecting section arranged on the inside is arranged on an inside edge section of the outer wall at which, in particular, the connecting face and a wall section face of the outer wall of one of the housing parts meet, for example for a connection forming the integration or for contacting the oppositely arranged housing part having the edge section.
(73) In particular, the protective section contacts an edge section, especially an edge thereof, of the outer wall of the oppositely arranged housing part.
(74) In particularly preferred embodiments, it is provided that at least one protective section is arranged on the outside along the connecting section. In particular, this is favourable so that contaminants are already prevented from entering at the outside and, for example, a sealing element arranged in the gap is also protected. In particular, the protective section provided on the outside is connected to one of the housing parts on the outside for integration thereon. In particular, the protective section provided on the outside contacts the oppositely arranged housing part on the outside, for example at an edge section and/or at an outer wall section face. In some advantageous embodiments, the protective section arranged on the outside is spaced from the oppositely arranged housing part.
(75) For example, the protective section arranged on the outside is arranged, for example connected or arranged in a contacting manner, at an outside edge section at which, in particular, the connecting face and an outer wall section face of the corresponding housing part meet.
(76) In some advantageous embodiments, the protective section arranged on the outside is arranged in a contacting or spaced-apart manner in relation to an outside wall section face.
(77) It is particularly advantageous if a protective portion at least partially covers the gap between the at least two housing parts along the connecting section. This offers, in particular, favourable protection against contaminants and/or for EMC shielding, since the area exposed by the gap is covered by the protective section. For example, the protective section is configured to cover the outside, whereby dirt in particular is already kept out at the outside. In some favourable embodiments, the protective section is configured to cover the inside, for example for EMC shielding or to protect the gap and/or interior on the inside.
(78) In particular, the protective section covers the gap transversely to its height extent, which runs between the limiting connecting faces. In particular, it is provided that the protective section extends at least between the geometric connection planes in which the connecting faces run and preferably at least on one side, preferably on both sides, i.e. projects beyond one or both of these geometric connection planes.
(79) In some particularly advantageous embodiments, it is provided that the at least two housing parts terminate flush on the outside, in particular in the region of the connecting section. In particular, this is favourable so that no disturbing projections occur at the transition of the housing parts and/or for a space-saving configuration of the connecting section.
(80) For example, a protective section is arranged on the outside and terminates flush with the oppositely arranged housing part. In particular, an outer wall section face and an outer face of the protective section transition flush into one another on the outside. In particular, the outside protective section is provided in particular in a contacting manner at an outside edge section.
(81) It is particularly favourable if at least one protective section abuts the oppositely arranged housing part under force. In particular, a spring-mounted protective section is in abutment under force. In particular, the force application allows a particularly favourable contacting and an

abutment that compensates for the tolerances of the oppositely arranged housing part.
(82) In some advantageous embodiments, it is provided that at least along the connecting section at least one protective section at least partially adapts to the oppositely arranged housing part by plastic deformation. In particular, this achieves good contacting and/or connection between the housing parts and thus a good protective effect.
(83) No further details have yet been given regarding other elements.
(84) It is particularly advantageous if, at least in the closed state, a sealing element is arranged between the two connecting faces of the at least two housing parts. In particular, the sealing element acts in a media-sealing manner, thus in particular against liquid and/or gaseous fluids and for example also against small particles. In particular, the interior is hereby sealed by the sealing element. Preferably, the sealing element runs continuously along the connecting section and in particular at least substantially along the connection contour.
(85) In particular, the sealing element is a separate part from the one protection section or plurality of protective sections, so that these can each be devised separately for their respective functions.
(86) The sealing element can be formed from a wide variety of sealing materials.
(87) For example, the sealing element is a metal sealing element.
(88) In other advantageous embodiments, the sealing element is made of a material comprising polymers.
(89) In particularly advantageous embodiments, it is provided that the sealing element is at least partially made of elastomer and is preferably an elastomer sealing element. For example, the sealing element is arranged as an elastomer sealing bead between the connecting faces.
(90) For example, in some embodiments, the sealing element is placed between the connecting faces during the assembly process.
(91) In particularly advantageous embodiments, it is provided that the sealing element is fixedly arranged on one of the housing parts, in particular on a housing part with a protective section, in particular is already fixedly arranged before the housing is assembled. In particular, this facilitates assembly and already offers a good sealing effect between the sealing element and the one housing part.
(92) In particular, the sealing element material is connected to at least one housing part in a force-locking and/or positively-locking manner.
(93) For example, the sealing element itself is formed from an adhesive material and is thus fixedly connected to the one housing part self-adhesively. In other advantageous embodiments, the sealing element is fixedly connected to the one housing part by means of an adhesive substance.
(94) It is particularly advantageous if at least one protective section, in particular for EMC shielding and/or for mechanical protection, is arranged on the inside relative to the sealing element, the inside protective section thus being arranged between the sealing element and the interior. Alternatively or additionally, it is particularly preferred if at least one protective section is arranged on the outside relative to the sealing element, the sealing element being arranged in particular between the outside protective section and the interior.
(95) Particularly preferred embodiments comprise an outside protective section, for example in the form of an embossed edge or in the form of a collar, in particular with recesses, the outside protective section favourably serving as mechanical protection for the sealing element, and an inside protective section for EMC shielding, which in particular at least approximately continuously contacts the opposite housing part, as well as a sealing element which is arranged between the two protective sections.
(96) It is particularly advantageous, in embodiments with a protective element, if the retaining section and the sealing element are arranged together on one of the housing parts, so that the same housing part is connected to the retaining section and the sealing element. For example, this simplifies construction and assembly of the housing parts, since only one of the housing parts needs to be devised for connection to the sealing element and the protective element.

(97) No further details have yet been given regarding other configurations of the housing and its housing parts.
(98) It is particularly advantageous if the at least two housing parts are pressed against one another. For example, this allows a stable connection between them and, in particular, the sealing element is acted upon by the pressure and its sealing effect is preferably increased.
(99) In particular, the at least two housing parts are pressed axially against each other. In particular, the axial pressing takes place at least approximately perpendicularly to a geometric plane in which one of the at least two housing parts, which is in particular lid-like, substantially extends, an extent of the latter perpendicularly to the geometric plane being in particular substantially smaller, for example at least five times, preferably at least ten times, smaller than its maximum extent in the geometric plane. It is particularly advantageous if the axial pressing is performed transversely, in particular at least approximately perpendicularly, to the connecting faces, in particular to geometric connection planes thereof. In particular, at least one outer wall of the at least two housing parts extends away from their connecting face at least approximately parallel to the axial pressing. Advantageously, assembly of the housing is simplified by the axial pressing, since in particular the housing parts only have to be substantially placed on top of each other and pressed.
(100) In particular, it is provided that at least substantially no radial pressing is provided between the at least two housing parts. In particular, a direction of the radial pressing runs at least approximately perpendicularly to a direction of the axial pressing. In particular, a wall section face extending away from the connecting face is or would be acted upon in the case of radial pressing. The absence of such a radial pressing simplifies in particular the assembly, and in particular distortions and/or creases in the housing parts are avoided.
(101) In particular, fastening means are provided which connect the two housing parts to each other at fastening points. In particular, the fastening points are provided on the outside of the connection contour in relation to the interior, so that the possibility of leakage into the interior is at least reduced or even completely prevented, particularly at these points.
(102) For example, flange projections are provided at the fastening points on the one, in particular lid-like, housing part and, for example, on the other of the at least two housing parts, its outer wall is reinforced, for example thicker, to accommodate the fastening means.
(103) It is particularly favourable if at least one protective section, for example at least on the outside, runs along the fastening points, and thus in particular the fastening points are also provided with appropriate protection.
(104) Preferably, the housing parts are acted upon by the fastening means.
(105) In particularly favourable embodiments, deformation limiters are arranged in the region of the fastening means. In particular, the deformation limiters limit an application of force by the fastening means and/or a deformation of the housing parts by the application of force and are configured to be correspondingly stable for this purpose.
(106) For example, the deformation limiters are transformed sections of one of the housing parts.
(107) It is particularly favourable if at least one protective section transitions into at least one deformation limiter. For example, at least one protective section in the region of at least one fastening point forms the deformation limiter, in particular with one or more of the explained features.
(108) Alternatively or in addition, the above-mentioned object is achieved by an electrotechnical apparatus which has a housing of the kind mentioned at the outset with at least one, preferably a plurality, of the features explained above, the advantages of these features being transformed to the electrotechnical apparatus.
(109) In particular, it is provided that at least one electrical component of the electrotechnical apparatus is arranged in the interior of the housing and, in particular, this and/or a surrounding area thereof is protected by the at least one integrated protective section as explained above.
(110) Furthermore, the object mentioned at the outset is alternatively or additionally achieved by a

motor vehicle component which has a housing and/or an electrotechnical apparatus with at least one, preferably a plurality, of the features explained above, the advantages described above being transformed to the motor vehicle component. In particular, in the case of the motor vehicle component, electromagnetic compatibility is important for an electrical component arranged in the interior of the housing and for the area surrounding the housing. In addition, the motor vehicle component in particular also requires mechanical protection, in order to keep dirt particles, for example created during operation of a motor vehicle, and/or fluids that act on the motor vehicle, for example weather-related liquids and/or cleaning liquids occurring during cleaning, for example by means of a high-pressure cleaner, in particular out of the interior of the housing.
(111) Furthermore, the object mentioned at the outset is alternatively or additionally achieved by a motor vehicle which has a housing and/or an electrotechnical apparatus and/or a motor vehicle component with at least one, preferably a plurality, of the features explained above, the advantages described above being transformed to the motor vehicle accordingly.
(112) Above and below, the wording is to be understood, at least approximately, in conjunction with a statement as meaning that technically conditional and/or technically irrelevant deviations are included in the statement, in particular that deviations of up to 20%, in particular of up to 10%, for example of up to 5%, are included. In the case of directional indications, deviations of up to 20°, in particular up to 10°, for example up to 5°, are included.
(113) The expressions “in particular”, “preferably”, “for example” and the like are to be understood to mean that the features mentioned in conjunction with these expressions are optionally but not necessarily provided in solutions in accordance with the invention.
(114) The above description of solutions in accordance with the invention thus comprises in particular the various combinations of features defined by the following consecutively numbered embodiments:
(115) 1. A housing (**100**), in particular for an electrotechnical apparatus, comprising at least two housing parts (**112**, **114**) which each have a connecting face (**126**, **156**), said connecting faces, in a closed state of the housing (**100**), facing one another along a connecting section (**196**), wherein at least one of the at least two housing parts (**112**, **114**), along the connecting section (**196**), has at least one integrated protective section (**212**, **232**), in particular for EMC shielding and/or for mechanical protection.
(116) 2. A housing (**100**) in accordance with embodiment 1, wherein exactly one protective section (**212**, **232**) is provided along the connecting section (**196**) or exactly two protective sections (**212**, **232**) are provided.
(117) 3. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is arranged captively on one of the housing parts (**112**, **114**).
(118) 4. A housing (**100**) in accordance with the preceding embodiments, wherein a section of the one housing part (**112**, **114**) forms at least one protective section (**212**, **232**).
(119) 5. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is a transformed section of the one housing part (**112**, **114**).
(120) 6. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is pressed or stamped into the one housing part (**112**, **114**).
(121) 7. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is a section integrally formed on the one housing part (**112**, **114**).
(122) 8. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is formed by a protective element (**352**) associated with one housing part (**112**, **114**), in particular the protective element (**352**) being connected to the one housing part (**112**, **114**) by an arrangement section (**354**).
(123) 9. A housing (**100**) in accordance with the preceding embodiment, wherein the protective element (**352**) is connected to the one housing part (**112**, **114**) by clinching and/or by a substance-to-substance bond, in particular welding.

(124) 10. A housing (**100**) in accordance with the two preceding embodiments, wherein the protective element (**352**) is formed from a metal material, in particular is a metal layer.
(125) 11. A housing (**100**) in accordance with the three preceding embodiments, wherein at least one protective section (**212**, **232**) is a transformed section of the protective element (**352**), in particular of the metal layer.
(126) 12. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is flat, in particular has a contacting face (**216**) or is formed flat to cover a gap between the connecting faces (**126**, **156**) in the closed state of the housing (**100**).
(127) 13. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) has an edge, in particular with a spike-like cross-section, which in particular protrudes beyond a connecting side (**124**, **154**) of the housing part (**112**, **114**) that has this protective section.
(128) 14. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is formed by a bead, in particular in one of the housing parts (**112**, **114**) and/or in the protective element (**352**).
(129) 15. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is spring-mounted.
(130) 16. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is resiliently connected to a section (**174**) forming the connecting face (**126**, **156**) and/or to the retaining section (**354**).
(131) 17. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**), in a closed state of the housing (**100**), contacts the oppositely arranged one of the at least two housing parts (**112**, **114**) along the connecting section (**196**) at least partially, in particular at least approximately continuously.
(132) 18. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**), in the closed state of the housing (**100**), at least partially abuts the connecting face (**126**, **156**) of the oppositely arranged housing part (**112**, **114**).
(133) 19. A housing (**100**) in accordance with the two preceding embodiments, wherein at least one protective section (**212**, **232**), in the closed state of the housing (**100**), at least partially abuts a wall section face (**148**, **149**) of the oppositely arranged housing part (**112**, **114**) extending away from the connecting face (**126**, **156**).
(134) 20. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) along the connecting section (**196**) is spaced at least partially, in particular continuously, from the oppositely arranged housing part (**112**, **114**), in particular from an outer wall section face (**149**) thereof.
(135) 21. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) along the connecting section (**196**) has one or more recesses (**372**), wherein the at least one protective section has two partial sections, in particular in the region of at least one recess (**372**), which run offset to each other transversely to the connection contour and the recess runs between the two partial sections in the region of the offset course.
(136) 22. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is arranged on the inside along the connecting section (**196**), in particular is arranged on an inside edge section (**144**) of an outer wall (**122**, **152**) of the housing (**100**).
(137) 23. A housing (**100**) in accordance with the preceding embodiments, wherein at least one protective section (**212**, **232**) is arranged on the outside along the connecting section (**196**), in particular at an outside edge section (**146**) and/or an outside wall section face (**149**) of an outer wall (**122**, **152**) of the housing (**100**).
(138) 24. A housing (**100**) in accordance with the preceding embodiments, wherein at least partially along the connecting section (**196**) at least one protective section (**212**, **232**) covers a gap (**188**)

between the at least two housing parts (**112**, **114**), in particular on the outside.
(139) 25. A housing (**100**) in accordance with the preceding embodiments, wherein the at least two housing parts (**112**, **114**) terminate flush on the outside, in particular at least one protective section (**212**, **232**) terminates flush with the opposite housing part.
(140) 26. A housing (**100**) in accordance with the preceding embodiments, wherein at least along the connecting section (**196**) at least one protective section (**212**, **232**) abuts the oppositely arranged housing part (**112**, **114**) under force.
(141) 27. A housing (**100**) in accordance with the preceding embodiments, wherein at least along the connecting section (**196**) at least one protective section (**212**, **232**) at least partially adapts to the oppositely arranged housing part (**112**, **114**) by plastic deformation.
(142) 28. A housing (**100**) in accordance with the preceding embodiments, wherein, at least in the closed state, a sealing element (**182**), in particular as a sealing bead, is arranged between the two connecting faces (**126**, **156**) of the at least two housing parts (**112**, **114**).
(143) 29. A housing (**100**) in accordance with the preceding embodiment, wherein the sealing element (**182**) is fixedly arranged on one of the housing parts (**112**, **114**).
(144) 30. A housing (**100**) in accordance with the two preceding embodiments, wherein at least one protective section (**212**, **232**) is arranged on the inside and/or outside relative to the sealing element (**182**).
(145) 31. A housing (**100**) in accordance with the three preceding embodiments, wherein the retaining section (**354**) and the sealing element (**182**) are arranged together on one of the housing parts (**112**, **114**), in particular the retaining section (**354**) is held between the one housing part (**112**, **114**) and the sealing element (**182**).
(146) 32. A housing (**100**) in accordance with the preceding embodiments, wherein the at least two housing parts (**112**, **114**) are pressed against one another, in particular axially.
(147) 33. An electrotechnical apparatus, wherein this comprises a housing (**100**) in accordance with the preceding embodiments, in particular at least one electrical component of the electrotechnical apparatus being arranged in the interior (**118**) of the housing.
(148) 34. A motor vehicle component which comprises a housing (**100**) and/or an electrotechnical apparatus in accordance with the preceding embodiments.
(149) 35. A motor vehicle which comprises a housing (**100**) and/or an electrotechnical apparatus and/or a motor vehicle component in accordance with the preceding embodiments.
(150) Further configurations and advantages of the invention are explained below by way of example in conjunction with several exemplary embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the drawing:
(2) FIG. **1** shows a plan view of a first exemplary embodiment of a housing;
(3) FIG. **2** shows an exploded depiction of the housing of the first exemplary embodiment;
(4) FIG. **3** shows a further exploded depiction of the housing of the first exemplary embodiment;
(5) FIG. **4** shows a sectional depiction of the housing of the first exemplary embodiment in accordance with the section IV-IV as shown in FIG. **1**;
(6) FIG. **5** shows a partial enlarged depiction of a region of a connecting section between two housing parts of the housing, said region being denoted by V in FIG. **4**;
(7) FIG. **6** shows a schematic sectional depiction of the connecting section of the first exemplary embodiment;
(8) FIG. **7** shows a schematic sectional depiction of a connecting section of a second exemplary embodiment;

(9) FIG. **8** shows a schematic sectional depiction of a connecting section of a third exemplary embodiment;
(10) FIG. **9** shows a schematic sectional depiction of a connecting section of a fourth exemplary embodiment;
(11) FIG. **10** shows a schematic sectional depiction of a connecting section of a fifth exemplary embodiment;
(12) FIG. **11** shows a schematic sectional depiction of a connecting section of a sixth exemplary embodiment;
(13) FIG. **12** shows a sectional depiction of a housing of a seventh exemplary embodiment;
(14) FIG. **13** shows a partial enlarged depiction of a region of a connecting section denoted by XIII in FIG. **12** with a fastening means;
(15) FIG. **14** shows a partial enlarged depiction of a region of the connecting section denoted by XIV in FIG. **12**;
(16) FIG. **15** shows a perspective depiction of a housing part, in particular a lid-like housing part, of the seventh exemplary embodiment, and
(17) FIG. **16***a*) and b) show a partial enlarged depiction of a variant of the housing part in accordance with FIG. **15** in the region of, in particular lid-like, overlapping protective sections.
DETAILED DESCRIPTION OF THE INVENTION
(18) An electrotechnical apparatus comprises a housing denoted as a whole by 100, which is shown by way of example in FIGS. **1** to **6**.
(19) The housing **100** comprises a plurality of housing parts, at least two housing parts **112** and **114**, which surround an interior **118** of the housing **100**. In particular, the interior **118** is a receptacle for one or more electrical devices, for example comprising electrical circuits and in particular comprising a car battery, which are not shown graphically in the figures.
(20) The housing part **112** comprises an outer wall **122** that at least partially surrounds the interior **118**. For example, the outer wall **122** surrounds the interior **118** largely laterally with, for example, four side wall sections.
(21) On a connecting side **124**, the outer wall **122** has a connecting face **126**.
(22) The connecting face **126** faces the other housing part **114** and runs in particular in a geometric connection plane **132**, up to which a part of the outer wall **122**, for example a connecting collar **128** thereof, extends.
(23) The connecting face **126** runs along a connection contour **134**, which here runs in the connection plane **132** and in a closed manner around a housing part opening **142**, so that the connecting face **126** frames the housing part opening **142**.
(24) The connecting face **126** extends transversely to the connection contour **134** between an inside edge section **144** and an outer edge section **146**, the inside edge section **144** in particular delimiting the housing part opening **142**, and the connecting face **126** and an inner wall section face **148**, which at least partially delimits the interior **118**, meeting at the inner edge section, for example in an edge-like manner. At the outer edge section **146**, the connecting face **126** meets, for example in an edge-like manner, with an outer wall section face **149** running on an outer side facing away from the interior **118**.
(25) In particular, the inner and outer wall section faces **148**, **149** extend obliquely, for example at least approximately perpendicularly, away from the geometric connection plane. For example, the connecting collar **128** provided extends between the inner and outer wall section faces **148** and **149** and is integrally formed on, but offset from, the remaining part of the outer wall **122**, in particular the side wall sections, so that their inner and outer side wall section faces are offset from the inner and outer wall section faces **148** and **149**. In variants, the inner and outer wall section faces **148**, **149** transition into the inner and outer side wall section faces respectively without being offset.
(26) For example, the outer wall **122** of the housing part **112** has further openings at which further housing parts can be arranged and/or through which parts of the electrotechnical apparatus, for

example cables, can be passed and the electrotechnical apparatus can be connected to other functional elements by means of these parts and is connected in a connected state. In other variants of the exemplary embodiment, it is provided that the outer wall **122** substantially surrounds the interior **118** in a closed manner and has only the one housing part opening **142**.
(27) The other housing part **114** also has an outer wall **152**, which, on a connecting side **154** thereof, has a connecting face **156** corresponding to the connecting face **126** of the housing part **112**.
(28) In particular, the connecting face **156** extends in a geometric connection plane **162** which, in an assembled state of the housing **100**, runs at least approximately parallel to the geometric connection plane **132** of the housing part **112**. In particular, the connecting face **156** also runs along a connection contour **164** which preferably has the same course as the connection contour **134** of the housing part **112**.
(29) The other housing part **114** is formed in particular in a lid-like manner, corresponding to the housing part opening **142** of the housing part **112**, at least one lid section **172** of the outer wall **152** substantially covering the housing part opening **142** in the closed state of the housing **100**, and a connecting part **174** of the outer wall **152** forming the connecting face **156** extending in a closed manner around the lid section **172**.
(30) In particular, the other housing part **114**, which is formed, for example, in a lid-like manner, is formed as a substantially flat wall part which extends substantially in two mutually perpendicular directions of extent and has a considerably smaller extent in a direction of extent running perpendicular to the two first-mentioned directions of extent. The two directions of extent preferably run at least approximately parallel to the geometric connection plane **162**.
(31) In addition, a sealing element **182** is provided on the connecting faces **126**, **156** along the connection contours **134**, **164** and is arranged captively in particular on one of the housing parts **112**, **114**, in this case on the, in particular lid-like, housing part **114**. For example, the sealing element **182** is fixedly connected to the housing part **112**, **114** adhesively.
(32) Preferably, the sealing element **182** is configured as an elastomer seal and is applied, for example, as a sealing bead to one of the housing parts **112**, **114**, in particular to its connecting face **126**, **156**.
(33) The sealing element **182** is intended for sealing media between the two housing parts **112**, **114** in the closed state and seals a gap **188** between the two connecting faces **126**, **156** of the two housing parts **112**, **114** media-tightly in the closed state and is thus configured in particular to seal against water, moisture and other liquids and, for example, against small solid particles, such as dirt, dust and/or abrasive particles.
(34) Thus, in the closed state of the housing **100**, in a connecting section **196**, sections of the two housing parts **112**, **114**, which form the connecting faces **126**, **156**, run correspondingly to one another along the connection contours **134**, **164**, and the connecting faces **126**, **156** of the latter face one another, and the gap **188** is formed therebetween and is closed at least approximately media-tightly by the sealing element **182**. In particular, the connecting section **196** runs circumferentially in a closed manner around the housing part opening **142**, which is covered by the cover section **172** in the closed state and is thus at least approximately closed media-tightly by the other housing part **114**, in particular together with the sealing element **182**.
(35) The at least two housing parts **112**, **114** are in particular formed from a metal material.
(36) In particular, the housing part **112** is a cast part in which the connecting face **126** is reworked, for example overmilled, and thus has lower tolerances than the non-reworked wall section faces **148**, **149**.
(37) The other housing part **114** is in particular made of a flat material, for example a sheet metal, and is preferably transformed into its final form.
(38) For EMC shielding, one of the housing parts **112**, **114**, in this exemplary embodiment the, in particular lid-like, housing part **114**, has a protective section formed as an integrated shielding

section **212**, which contacts the opposite housing part **112** along the connecting section **196** at least in sections and preferably completely along the connecting section **196**. In this exemplary embodiment, the shielding section **212** is a transformed section of the housing part **114**, in particular of its outer wall **152**.
(39) The shielding section **212** comprises a contacting face **216**, which abuts opposite a housing part **112** and contacts the inside edge section **144**, for example an edge thereof, in particular under application of a force.
(40) The shielding section **212** runs slightly offset here on the interior side from the connection contour **164** and is arranged between the cover section **172** and the section of the outer wall **152** forming the connecting face **156** and connects these sections.
(41) In particular, the shielding section **212** is integrally formed on the section with the connecting face **156** and extends obliquely with respect to the latter and the geometric connection plane **162**, for example at an angle between 30° and 70°, away from the latter in the direction of the opposite housing part **112** and in particular to such an extent that, in the closed state of the housing **100**, the shielding section **212** protrudes beyond the geometric connection plane **132** of the other connecting face **126** and engages, at least with a small section, in the housing part opening **142**.
(42) On a side of the shielding section **212** which is arranged opposite its side facing the section **174** with the connecting face **156**, the shielding section **212** is in particular integrally formed on the cover section **172**, and in the closed state in particular the cover section **172** runs at least approximately parallel to the geometric connecting face **132** and is arranged, for example, laterally of the connecting collar **128** at least partially in the housing part opening **142**, and the shielding section **212** is integrally formed on the cover section **172** by the region of the shielding section that engages in the housing opening **142**.
(43) In this case, the contacting face **216** is arranged on the same side of the outer wall **152** as the connecting face **156** and an inner face of the cover section **172** facing the interior **118**.
(44) Preferably, the housing part **114** has a further protective section formed as a cover section **232**, which at least partially covers the gap **188** along the connecting section **196** on an outer side **236** averted from the interior **118**.
(45) Preferably, the cover section **232** is formed as a transformed section of the, in particular lid-like, housing part **114**.
(46) In the closed state, the cover section **232** extends at least between the two geometric connection planes **132** and **162** and preferably projects beyond them on both sides. In particular, the cover section **232** is integrally formed on the section **174** with the connecting face **156** and extends away therefrom in the direction of the oppositely arranged housing part **112**, wherein in particular an end region **238** of the protective section **232** is arranged overlapping on the outside with a section of the housing part **212**, in particular of the connecting collar **128**, in front of at least a part of the outer wall section face **149**.
(47) In some variants, the cover section **232**, in particular with its end region **238**, at least partially contacts the opposite housing part **112**, for example at the outer wall section face **149** thereof, whereby in particular the EMC shielding is improved. In other variants, a small gap is provided in particular in the overlapping region between the cover section **232**, in particular the end region **238** thereof, and the opposite housing part **112**, in particular the outer wall section face **149**, whereby larger tolerances on the cast housing part **112** can be better compensated and assembly of the housing parts **112**, **114** is simplified and potential stresses are at least reduced.
(48) The housing parts **112** and **114** are connected to each other by means of fastening means **252** at fastening points **254**. In particular, the housing parts **112** and **114** are screwed together by means of screws as fastening means **252** at fastening points **254** having screw holes.
(49) Preferably, the fastening points **254** are arranged relative to the interior **118** outwardly of the connection contours **134** and **164**, and in this variant of the exemplary embodiment the fastening points **254** are arranged between the connection contours **134** and **164** on the one hand and the

contour along which the cover section **232** runs on the other hand.
(50) For example, the housing part **112**, in particular at its connection collar **128**, has wall thickenings, which are provided with screw holes.
(51) The other, in particular lid-like, housing part **114** has, for example, flange projections at the fastening points **254**, said flange projections being provided in particular with apertures for the fastening means **252**. Preferably, the flange projections are provided outwardly with partial sections of the cover section **232**.
(52) In particular, by means of the fastening means **252**, the two housing parts **112**, **114** are pressed at least slightly against each other in the closed state, so that the sealing element **182** is compressed between them both, in particular between the two connecting faces **126** and **156**, and thus has a sealing effect in the gap **188** in a particularly favourable manner.
(53) In addition, the shielding section **212** is applied under force against the oppositely arranged housing part **112** so that, due to the application of force and a flexible inherent elasticity of the shielding section **212**, the latter contacts the oppositely arranged housing part **114** at least approximately completely along the connecting section **196**, even with large tolerances.
(54) In a second to sixth exemplary embodiment, shown by way of example in FIGS. **7** to **11**, a housing **100** comprises at least two housing parts **112**, **114** which at least partially surround an interior **118** and have respective sections along a connecting section **196** which face each other in a closed state of the housing. For example, the housing **100** with the at least two housing parts **112**, **114** is formed at least substantially as in the first exemplary embodiment and has one or more of the features explained in connection with the first exemplary embodiment, in particular with respect to the connecting faces **126**, **156** and/or the outer walls **122**, **152** and/or the geometric connection planes **132**, **162** and/or the connecting collar **128** and/or the cover section **172**. In particular, in these exemplary embodiments, a sealing element **182**, for example with one or more of the features explained above in conjunction with the first exemplary embodiment, is also provided.
(55) Elements that are of at least substantially identical configuration and/or fulfil at least the same basic function are previously and subsequently provided with the same reference sign. In particular, if special reference is to be made to an alternative variant in an exemplary embodiment, a letter referring to the corresponding exemplary embodiment is appended as a suffix to the reference sign.
(56) In the second exemplary embodiment, which is shown schematically in FIG. **7**, the housing part **114**, which in particular is lid-like, has at least one protective section in the form of a shielding section **212***a*, which is embossed in the housing part **114**.
(57) The embossed protective section, in this case the shielding section **212***a*, has an embossing part **312** protruding from the connecting side **154** and in particular beyond the geometric connection plane **162** and extending in the direction of the oppositely arranged housing part **112**. In particular, the embossing part **312** is produced by embossing through the housing part **114**, so that a side of the housing part **114** opposite the connecting side **154** has a groove **314** running opposite the embossing part **312** and formed in a manner corresponding thereto.
(58) At its embossing part **312**, the embossed section has a contacting face **216***a*, which faces the oppositely arranged housing part **112** and run, for example, at least approximately parallel to the geometric connection plane **162** and substantially along the connection contour **164**. In the closed state, the contacting face **216***a* of the embossing part **312** contacts the oppositely arranged housing part **122** at least approximately completely along the connecting section **196**, in particular at the inside edge section **144**, the contacting face **216***a* preferably abutting at least partially on the connecting face **226**, in particular under the application of a force.
(59) Preferably, the embossed section is resilient, so that its spring effect contributes to a consistently good contact.
(60) For example, the embossed section is formed as a bead.
(61) Preferably, the embossed section **212***a* is arranged on the interior side and is supported, for

example, on the inside edge section **144**. In particular, the embossed section thus runs along the connecting face **156** and between the section **174**, which has the connecting face **156**, and the cover section **172** of the housing part **114**.
(62) It is particularly favourable if, in this exemplary embodiment, the housing part **114** also has an outside protective section formed as a cover section **232**. This is preferably formed at least substantially as explained in conjunction with the first exemplary embodiment, and therefore full reference is made to those explanations.
(63) In a third exemplary embodiment, which is shown schematically in FIG. **8**, the housing part **114**, which is in particular lid-like, has a protective section which is formed as a combined shielding section **212***b* and cover section **232***b* and is provided on the outer side averted from the interior **118**.
(64) In some variants, this outside section is configured as a sheared-off edge of the housing part, in particular its outer wall **172**. In other variants of the exemplary embodiment, this protective section is formed as a transformed section of the housing part **114**, in particular its outer wall **172**.
(65) A part of the protective section **212***b*, **232***b* protrudes from the connecting side **154** of the housing part **114**, in particular beyond the geometric connection plane **162**, and extends in the direction of the oppositely arranged housing part **112**, which this section **212***b*, **232***b* preferably contacts at least approximately continuously along the connecting section **196**.
(66) The protective section **212***b*, **232***b* has a contacting face **216** at which it contacts the oppositely arranged housing part **112**. In the variant shown in FIG. **8** by way of example, the protective section **212***b*, **232***b* abuts with the contacting face **216** on at least part of the connecting face **126** of the oppositely arranged housing part **112**, in particular under the application of a force.
(67) In particular, the contacting face **216** of the protective section **212***b*, **232***b* abuts the outer edge section **146**, for example at an outer edge thereof. In other variants, it is provided that the outside protective section **212***b*, **232***b* has an inwardly directed contacting face which contacts the outer wall section face **149** of the oppositely arranged housing part **112**.
(68) In particular, in the closed state the protective section **212***b*, **232***b* extends from its housing part **114** beyond its geometric connection plane **162** to at least the geometric connection plane **132** of the oppositely arranged housing part **112** and thus covers the gap **188** between the connecting faces **126** and **156** from the outside.
(69) In this exemplary embodiment, too, a sealing element **182** is provided which, in the closed state, acts in a sealing manner between the connecting faces **126** and **156**, and which is arranged relative to the protective section **212***b*, **232***b* on the inside towards the interior **118**. In particular, the sealing element **182** is formed at least substantially the same as in the exemplary embodiments explained above, and therefore full reference is made to the explanations above with regard to advantageous configurations of said sealing element.
(70) In a fourth exemplary embodiment, which is shown schematically by way of example in FIG. **9**, the housing part **114**, which is in particular lid-like, has a protective section which is configured as a shielding section **212***c* and has a bevelled edge **332**, the edge **332** being pressed against the opposite housing part **112**, in particular into the connecting face **126**, in the closed state.
(71) In particular, the section with the edge **332** on the connecting side **154** protrudes here beyond the connecting face **156** with an extent which is at least as great as a height extent of the gap **188** measured from one to the other housing part **112**, **114** in the closed state. In particular, the extent of this spacing is at least as great as the spacing between the geometric connection planes **162**, **132** in the closed state.
(72) The section with the edge **332**, said section being spike-like in cross-section, for example, has two apex faces **334**, **336** which converge towards each other at the edge **332** and meet at a point at a protruding end of the edge **332** in an edge tip **338**.
(73) In the closed state, the protective section **212***c* thus extends from its housing part **114** to the opposite housing part **112** and is pressed onto the latter with the edge tip **338** and, in particular, is

pressed into the opposite housing part with the edge tip **338** in the region of the connecting face **126**. Thus, in the closed state, the section **212***c* having the edge bridges the gap **188** and preferably contacts the opposite housing part **112** at least approximately completely along the connecting section **196**.
(74) In this exemplary embodiment, the section **212***c* is arranged on the inside relative to the sealing element **182** and contacts the inside edge section **144**. In other variants of the exemplary embodiment, it is provided that this section **212***c* is arranged with the edge **332** on the outside relative to the sealing element **182**, for example contacting the outer edge section **146**, and thereby alternatively or additionally acts as a cover section.
(75) Preferably, in the exemplary embodiment, in particular in the variant with the inside section **212***c* with the edge **332**, an outer protective section formed as a cover section **232** is additionally provided. In particular, this outer protective section is formed at least substantially as in one of the exemplary embodiments explained above with one or more of the features explained therein, and therefore full reference is made to the preceding explanations.
(76) In a fifth exemplary embodiment, schematically shown in FIG. **10** by way of example, the integrated protective section is formed by a shielding section **212***d* of a protective element **352** which is connected to the housing part **114** and which is preferably a metal layer.
(77) In this case, the metal layer has a retaining section **354**, at which it is connected to the housing part **114** on the connecting side **154**.
(78) In particular, the retaining section **354** is flat and abuts the connecting face **156**.
(79) For example, the protective element **352** is connected to the housing part **114** merely by the application of the retaining section **354** by the sealing element **182**, this being achieved in particular by the fastening means **252**, although, in preferred variants of the exemplary embodiment, the protective element **352** is arranged captively on the, in particular lid-like, housing part **114**, for example by welding, in particular by spot welding, already before the housing **100** is assembled.
(80) The shielding section **212***d* is integrally formed on the retaining section **354** and is a transformed section of the metal layer forming the protective element **352**.
(81) In particular, the shielding section **212***d* is a section protruding obliquely from the retaining section **354**.
(82) The shielding section **212***d* extends from an end region abutting the housing part **114** away from the connecting side **154** of the housing part **114** to a second end region, with the extent of the shielding section **212***d* away from the housing part **114** being at least of such a magnitude that the second end region is remote from the connecting side **154** by at least the height extent of the gap **188** in the closed state.
(83) The shielding section **212** has a contacting face **216** facing the opposite housing part **112** and contacts the housing part **112** at said contacting face at least approximately completely along the connecting section **196**. For example, the protective element **352** is connected to the housing part **114** in such a way that, in the closed state, the contacting surface **216** of the protective element contacts the inside edge section **144**, in particular an edge thereof, of the opposite housing part **112**. In this case, it is favourable if the shielding section **212***d*, in the closed state, abuts the housing part **114** within the gap **188** and extends obliquely to the direction of the height extent of the gap **188** at least as far as the oppositely arranged housing part **112** and projects with an end region out of the gap **188** and, in so doing, contacts the opposite housing part **112** with the contacting face **216** facing the latter, in particular by applying force resiliently to said contacting face.
(84) Preferably, the shielding section **212***d* is deformed, in particular elastically deformed, by the loading on the housing parts **112**, **114** compared to an unloaded state.
(85) The shielding section **212***d* is arranged on the interior side in the gap **188** and is thus arranged between a sealing element **182**, arranged in the gap **188**, and the interior **118**. In variants of the exemplary embodiment, the shielding section **212***d* is correspondingly arranged on the outside, in particular contacting the outer edge section **146**.

(86) It is particularly favourable if the sealing element **182**, on the one hand, sealingly abuts the connecting face **126** of the housing part **112** and, on the opposite side, partially sealingly abuts the connecting face **156** of the other housing part **114** and the retaining section **354** of the protective element **352**.
(87) Preferably, the housing part **114** further has a cover section **232**, which in particular is formed at least substantially as in one of the exemplary embodiments explained above and in particular has one or more of the features explained in conjunction therewith, and therefore in this regard full reference is made to the explanations above.
(88) In a sixth exemplary embodiment, which is shown schematically by way of example in FIG. **11**, a protective element **352***e* forming the integrated protective section is connected to the housing part **114**, in particular by means of clinching. The protective element **352***e* is preferably formed from a flat metal material.
(89) In this case, a retaining section **354***e* of the protective element **352***e* abuts a joining section **362** of the housing part **114** on the connecting side **154** thereof, and in this region the retaining section **354***e* and the joining section **362***e* are at least partially connected to one another in a force-locking and/or positively-locking manner.
(90) In particular, the joining section **362** is a transformed section of the housing part **114**, the latter having material accumulations **364** at joining points on the connecting side **154** which bulge out from the connecting side **154** and form the connection to the retaining section **354***e*, and in particular having corresponding channels at the joining points on a side opposite the connecting side **154**. In particular, the retaining section **354***e* has apertures through which the material accumulations **364** engage and are surrounded on both sides in order to produce the captive connection.
(91) For example, the joining section **362** is part of the cover section **172**. In other favourable variants of the exemplary embodiment, the joining section **362** is arranged between the cover section **172** and the section **174** of the housing part **114** forming the connecting face **176**.
(92) Furthermore, the protective element **352***e* has a protective section, which is in the form of a shielding section **212***e* and in particular abuts with a contacting face **216***e* over the entire surface on the connecting face **126** of the oppositely arranged housing part **112**, in particular on the inside edge section **144**. In particular, the retaining section **354***e* and the shielding section **212***e* are connected to each other by means of a bridging section **366** bridging the spacing between the housing parts **114** and **112**. Preferably, the retaining section **354***e* and the shielding section **212***e* extend at least approximately parallel to each other and the bridging section **366** extends obliquely to these sections **354***e*, **212***e*, between them, and transitions into them at edge sections. In particular, the protective element **352** thus has a half bead along the connecting section **196** in this exemplary embodiment, with the bead feet forming the edge sections.
(93) It is particularly favourable if the retaining section **354***e* and the shielding section **212***e* have a greater spacing, which is measured in particular at least approximately perpendicularly to their substantially parallel extent, in a state with clearance than in the closed state of the housing **100**, so that this region of the protective element **352**, preferably at the bridging section **366**, is in particular elastically deformed in the closed state and abuts the housing parts **114** and **112** in a loading manner.
(94) The protective element **352***e* is arranged on the interior side between the housing parts **114** and **112** and at least the shielding section **212***e* projects into the gap **188**. Thus, the protective element **352***e* is arranged between the interior **118** and a sealing element **182** arranged in the gap **188**.
(95) In the closed state of the housing **100**, the sealing element **182** sealingly abuts the connecting faces **126** and **156** of the two housing parts **112**, **114**, and in some variants of the exemplary embodiment also abuts the protective element **352***e*, in particular the shielding section **212***e*.
(96) The sealing element **182** is in particular configured at least substantially as in the exemplary embodiments explained above and in particular has one or more of the features explained above,

and therefore in this regard full reference is made to the above explanations.

(97) In addition, the housing part **114** preferably has a protective section which is in the form of a cover section **232** and is arranged on the outside and, in particular, is formed at least substantially as in the exemplary embodiments explained above and preferably has one or more of the features explained therein, and therefore in this regard full reference is made to the above explanations.

(98) In all other respects, unless otherwise explained and/or explained in more detail in conjunction with one of the preceding exemplary embodiments, elements and features are configured as in another of the exemplary embodiments explained above, in particular as in exemplary embodiment 1, and therefore full reference is made to the respective explanations.

(99) In particular, for example inside and outside protective sections, in particular cover sections and shielding sections, which have been explained in conjunction with different exemplary embodiments, can be combined with each other.

(100) A seventh exemplary embodiment of a housing denoted as a whole by 100, shown by way of example in FIGS. **12** to **15**, comprises at least two housing parts **112**, **114**, which again define an interior **118**.

(101) The first housing part **112** again comprises an outer wall **122**, which extends at a connecting side **124** to a connecting face **126**. In particular, the connecting face **126** extends between an inside edge section **144** facing the interior **118** to an edge section **146** running on an opposite side, which is in particular an outer side of the housing. In particular, the connecting face **126** extends substantially in a geometric connection plane **132**.

(102) The outer wall **122** extends away from the connecting face **126** and has an inside wall section face **148** delimiting the interior **118** and, on an opposite side, an outer wall section face **149**, which run at least approximately parallel to one another, for example in a connection end region arranged on the connecting side **124**, and extend away at least approximately perpendicularly from the geometric connection plane **132**.

(103) At the inside edge section **144**, the inside wall section face **148** abuts the connecting face **126**, and at the outside edge section **146** the outer wall section face **149** abuts the connecting face **126**, particularly in an edge-like manner.

(104) The connecting face **126** runs along a connection contour **134** and delimits a housing part opening **142** which adjoins the inside edge section **144** on the interior side, and, in particular, the connection contour **134** and thus the connecting face **126** run circumferentially in a closed manner around the housing part opening **142**.

(105) For example, the housing part **112** is a cast part, in particular with a milled connecting face **126**.

(106) The second housing part **114** also has an outer wall **152**, which at least partially delimits the interior **118** and, in particular, in a closed state of the housing, closes the housing part opening **142** in a lid-like manner. This outer wall **152** of the housing part **114** also has, on a connecting side **154**, a connecting face **156** which runs along a connection contour **164**, the connection contour **164** running correspondingly to the connection contour **134** of the oppositely arranged housing part **112**, so that the two connecting faces **156**, **126** of these housing parts **114**, **112** face one another along a connecting section **196** in the closed state of the housing.

(107) With a cover section **172**, the outer wall **152** in the closed state delimits the housing part opening **142** of the oppositely arranged housing part **112**, wherein the cover section **172** is in particular circumferentially framed by the section **174** with the connecting face **156**.

(108) In particular, this housing part **114** is formed substantially as a flat housing part **114** which extends substantially parallel to a geometric connection plane **162** in which, in particular, the connecting face **156** runs, and an extent of this flat housing part **114** perpendicular to the geometric connection plane **162** is substantially less than the extent of said flat housing part parallel to the geometric connection plane **162**.

(109) Preferably, the housing part **144**, which is in particular lid-like, is formed from a metal

material, in particular a sheet metal.

(110) In the closed state of the housing **100**, there is a gap **188** between the two connecting faces **126**, **156** of the housing parts **112**, **114**, the height dimension of said gap corresponding substantially to a spacing between the two connecting faces **126**, **156**.

(111) To seal the gap **188**, a sealing element **182** is provided therein and extends in particular along the connection contours **134**, **164** along the connecting section **196** and sealingly abuts the connecting faces **126** and **156** on one side each.

(112) For example, the sealing element **182** abuts flat sections of the connecting faces **126**, **156**. In variants, a groove, in which the sealing element **182** is partially received, is provided on a connecting face **126**, **156**, in particular the connecting face **126** of the cast housing part **114**.

(113) For protection, in particular against external influences, the housing part **114**, which in particular is lid-like, has a protective section which runs in particular circumferentially at the edge and is formed as a cover section **232***f*.

(114) Preferably, sections of the cover section **252***f* are also arranged at the fastening points **254** around the apertures for the fastening means **252** and frame the various apertures at least at the outside and partially completely. In particular, the fastening points **254** on the lid-like housing part **114** are formed as flange projections.

(115) In this case, this cover section **232***f* does not run circumferentially in a closed manner along the connection contour **164**, but extends only in sections along this contour **164** along the connecting section **196**. Recesses **372** are provided between the individual sections of the cover section **232***f*.

(116) In particular, the recesses are provided in the region of the fastening points **254**, as for example in the variant of the exemplary embodiment shown in FIG. **15**. In this case, the recesses are arranged in each case between a section framing a corresponding aperture at least at the outside and a section of the cover section **232***f* running in particular continuously between two adjacent fastening points. Preferably, these recesses are large enough, for example, to allow moisture to escape and to avoid a capillary effect, but as small as possible to keep an interruption of the protection as low as possible.

(117) For example, in an adjoining region **376**, in which another component adjoins the housing on the outside, the outer cover section **232***f* is also interrupted with a recess, since in this region external protection, in particular mechanical protection, is already provided by the adjoining component.

(118) In variants of the exemplary embodiment, as shown by way of example in FIG. **16***a*) and b), at least some recesses are formed in a twisted manner.

(119) Here, two partial sections **382** and **384** of the protective section **232***f* run in the region of the recess **372′** substantially along the connection contour **164** but offset to each other transversely thereto. Thus, a free space of the recess **372′** in the region of the offset course of the two partial sections **382**, **384** runs substantially along the connection contour **164**, specifically substantially from one end **386** of one partial section **382** to one end **384** of the other partial section **388**. At each of the ends **386** and **388**, the recess **372′** opens to one transverse side each of the protective section **232***f*, with one transverse side here facing the sealing element **182** and the other transverse side facing outwards. Thus, also in the region of the recess **372′**, the protective section **232***f* does not have a free passage running at least approximately perpendicularly to its longitudinal extent or at least approximately perpendicularly to the connection contour **164**, and therefore preferably also in this region the sealing element arranged further inwards is protected from external damaging effects, such as liquid jets of a high-pressure cleaner during cleaning.

(120) For example, the angled recesses **372′** are provided between two fastening points and preferably in the region of a fastening point **254**, so that at least one of the sub-sections, for example the sub-section **384**, is formed by a section of the protective section **232***f* at least partially surrounding an aperture.

(121) The sections of the cover section **232***f* rest on the connecting face **126** of the oppositely arranged housing part **112**, specifically on the outside, so that the sealing element **182** is arranged between the interior **118** and the cover section **232***f*. In particular, the sections of the cover section **232***f* abut on the outer edge section **146** and are flush with the outer wall **122**, in particular the outer wall section face **149** of the oppositely arranged housing part **112**.
(122) In this exemplary embodiment, the cover section **232***f* is formed as a transformed section of the housing part **114**, in particular the sections thereof are embossed sections.
(123) In variants of the exemplary embodiment, the sections of the cover section **232***f* are alternatively or additionally formed with features explained in conjunction with the protective sections in the other exemplary embodiments.
(124) In another variant, the cover section **232***f* runs continuously along the connecting section **196**, for example circumferentially in a closed manner around the housing part opening **142**.
(125) Preferably, a protective section in the form of a contacting section **212** is also provided on the inside in the case of this housing **100** and in particular contacts the inside edge section **144**. With regard to the configuration of this contacting section, full reference is made to the preceding explanations for the exemplary embodiment with the embossed contacting section. In variants of the exemplary embodiment, the contacting section **212** is formed as in another of the exemplary embodiments described above, and therefore in respect of these variants full reference is made to the explanations above.
(126) Furthermore, features which have been explained in conjunction with this seventh exemplary embodiment, in particular with regard to the housing and its at least two housing parts **112**, **114** and/or the protective section which is formed as a cover section **232***f* and which in particular terminates flush with the oppositely arranged housing part, can also be transformed to the other exemplary embodiments explained above and/or combined with features and configurations which have been explained in conjunction with these preceding exemplary embodiments.
(127) Moreover, features and elements not explained in greater detail in one exemplary embodiment are preferably formed as in another of the exemplary embodiments explained above, and therefore full reference is made to the corresponding explanations.

Claims

1. A housing comprising at least two housing parts which each have a connecting face, said connecting faces, in a closed state of the housing, facing one another along a connecting section, wherein at least one of the at least two housing parts, along the connecting section, has at least one integrated protective section, wherein at least partially along the connecting section, the at least one integrated protective section externally covers a gap between the at least two housing parts, wherein the at least two housing parts surround an interior of the housing and abut each other at an abutment in a geometric connection plane, wherein the gap extends radially outward to the at least one integrated protective section, wherein the at least one integrated protective section extends axially to the geometric connection plane or beyond the geometric connection plane so as to externally cover the gap.
2. The housing in accordance with claim 1, wherein at least one protective section is provided along the connecting section for at least one of EMC shielding and/or mechanical protection.
3. The housing in accordance with claim 1, wherein at least one protective section is arranged captively on one of the housing parts.
4. The housing in accordance with claim 1, wherein a section of the one housing part forms at least one protective section.
5. The housing in accordance with claim 1, wherein at least one protective section is a transformed section of the one housing part.
6. The housing in accordance with claim 1, wherein at least one protective section is a section

integrally formed on the one housing part.
7. The housing in accordance with claim 1, wherein at least one protective section is formed by a protective element associated with one housing part.
8. The housing in accordance with claim 7, wherein the protective element is connected to the one housing part by at least one of clinching and/or a substance-to-substance bond.
9. The housing in accordance with claim 7, wherein the protective element is formed from a metal material.
10. The housing in accordance with claim 1, wherein at least one protective section has an edge, which protrudes beyond a connecting side of the housing part that has this protective section.
11. The housing in accordance with claim 1, wherein at least one protective section is formed by a bead.
12. The housing in accordance with claim 1, wherein at least one protective section is spring-mounted.
13. The housing in accordance with claim 1, wherein at least one protective section is resiliently connected to a section forming at least one of a connecting face and/or a retaining section.
14. The housing in accordance with claim 1, wherein at least one protective section, in the closed state of the housing, at least partially abuts the connecting face of the oppositely arranged housing part.
15. The housing in accordance with claim 1, wherein at least one protective section, in the closed state of the housing, at least partially abuts a wall section face of the oppositely arranged housing part, the wall section face extending away from a connecting face.
16. The housing in accordance with claim 1, wherein at least one protective section along the connecting section is spaced at least partially from the oppositely arranged housing part.
17. The housing in accordance with claim 1, wherein at least one protective section along the connecting section has one or more recesses, wherein the at least one protective section has two partial sections which run offset to each other transversely to the connection contour and the recess runs between the two partial sections in the region of the offset course.
18. The housing in accordance with claim 1, wherein at least one protective section is arranged on the inside along the connecting section.
19. The housing in accordance with claim 1, wherein at least one protective section is arranged on the outside along the connecting section.
20. The housing in accordance with claim 1, wherein at least along the connecting section at least one protective section abuts the oppositely arranged housing part under force.
21. The housing in accordance with claim 1, wherein at least along the connecting section at least one protective section at least partially adapts to the oppositely arranged housing part by plastic deformation.
22. The housing in accordance with claim 1, wherein at least in the closed state, a sealing element is arranged between the two connecting faces of the at least two housing parts.
23. The housing in accordance with claim 22, wherein at least one protective section is arranged on the inside and/or outside relative to the sealing element.
24. The housing in accordance with claim 22, wherein a retaining section and the sealing element are arranged together on one of the housing parts, and the retaining section is held between the one housing part and the sealing element.
25. An electrotechnical apparatus, which comprises a housing in accordance with claim 1, wherein at least one electrical component of the electrotechnical apparatus being arranged in an interior of the housing.
26. A motor vehicle component which comprises a housing in accordance with claim 1.
27. A motor vehicle which comprises at least one of a housing in accordance with claim 1.
28. A motor vehicle which comprises an electrotechnical apparatus in accordance with claim 25.
29. A motor vehicle which comprises a motor vehicle component in accordance with claim 26.

30. The housing in accordance with claim 1, wherein the at least two housing parts abut each other at the abutment directly or via a protective element.